HW#1

Friday, 6 February 2026 10:53 AM

Q1] Effusion of gas through a small hole

$f(\vec{v})\,d\vec{v}$ = probability of a given molecule having a velocity vector b/w $\vec{v}$ and $\vec{v}+d\vec{v}$

$$f(\vec{v}) = f(|\vec{v}|) = f(v^2) \quad ; \quad \iiint f(\vec{v})\,d^3v = 1$$

$$f(v^2) = C\,e^{-av^2} \quad ; \quad a>0$$

(Maxwell Distribution)

$$\langle v_x \rangle = \bar{v}_x = \int_{-\infty}^{\infty} v_x f_x(v_x^2)\,dv_x$$

$$\langle v_x^2 \rangle = \overline{v_x^2} = \int_{-\infty}^{\infty} v_x^2 f_x(v_x^2)\,dv_x$$

$$\int_{-\infty}^{\infty} f(v^2)\,dv = 1$$

Solving this in spherical coordinates gives

$$\langle v \rangle = \bar{v} = 4\pi \int_0^{\infty} v\, f(v^2)\, v^2\,dv$$

$$\langle v^2 \rangle = \bar{v}^2 = 4\pi \int_0^{\infty}$$

Number of molecules crossing the hole per unit time w/ speed v is $\bar{v} f_E(\bar{v})\,d\bar{v}$

$$\langle v_E \rangle = \frac{\int_0^{\infty} \bar{v}\, f_E(\bar{v})\,d\bar{v}}{\int_0^{\infty} f_E(\bar{v})\,d\bar{v}}$$

↓

solving this integral using $f_E(\bar{v})\,d\bar{v} = \left(\frac{m}{2\pi kT}\right)^{1/2} e^{-m\bar{v}^2/2kT}\,d\bar{v}$ from Maxwell-Boltzmann dist.

& some math...

Similarly, $\langle v_E^2 \rangle = \int_0^{\infty} \bar{v}^2 f_E(\bar{v})\,dv$ this should yield $2kT$

Similarly, $\langle v_E^2 \rangle = \dfrac{\int_0^\infty \bar{v}^2 f_E(\bar{v})\, d\bar{v}}{\int_0^\infty f_E(\bar{v})\, d\bar{v}}$ → this should yield $2kT$

Q2] Maxwell-Boltzmann for a 2D gas ; v_x and v_y direction

Probability of speed for 2D ideal gas b/w $\bar{v}$ and $\bar{v}+d\bar{v}$ → $f_{2D}(\bar{v})\, d^2\bar{v} = \int_0^{2\pi} f(v_x, v_y)\, d^2v$

From Boltzmann dist., $f(v_x, v_y) = \dfrac{m}{2\pi kT} e^{-m(v_x^2+v_y^2)/2kT}$

$\bar{v}^2 = v_x^2 + v_y^2$ (Ideal gas)

Q6] Random walk in 3D

N → total number of steps during the walk

(a) $\langle R \rangle$, $\langle R^2 \rangle$ vs. N

↑ Average length

(b) $P(R) = 4\pi R^2 \left(\dfrac{3}{2\pi N}\right)^{3/2} \exp\left(\dfrac{-3R^2}{2N}\right)$

↳ probability density that the random walk has an end-to-end distance equal to R

What happens as $N \to$ large